

Introduction to Contract Theory and Auctions

Knowledge and Beliefs

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Definition 1.1

A **knowledge space** is a tuple $(N, \Omega, \{T_i\}_{i \in N}, \{\tau_i\}_{i \in N})$ where

- 1 N is the set of players.
- 2 Ω is the (*finite*) set of **states of the world**.
- 3 T_i is the space of possible **types** of player i .
- 4 $\tau_i : \Omega \rightarrow T_i$ is player i 's **private signal** or **type**.

Definition 1.2

Player i 's **information partition function** $P_i : \Omega \rightarrow 2^\Omega$ is defined by

$$P_i(\omega) = \{\omega' : \tau_i(\omega') = \tau_i(\omega)\} = \tau_i^{-1}(\tau_i(\omega)).$$

The collection $\mathcal{P}_i = \{P_i(\omega)\}_{\omega \in \Omega}$ is called **i 's information partition**.

Remark

Knowledge spaces can be also given in terms of partitions, i.e., $N(\Omega, \{\mathcal{P}_i\}_{i \in N})$.

Partitions

A partition of Ω is a family $\mathcal{P}$ of pairwise-disjoint non-empty subsets of Ω whose union includes Ω . That is, a partition $\mathcal{P}$ satisfies

P1 $\bigcup \mathcal{P} = \bigcup_{P \in \mathcal{P}} P = \Omega.$

P2 $A, B \in \mathcal{P}, A \cap B \neq \emptyset \Rightarrow A = B.$

Example 1.3

Ω is the set of all possible outcomes of tossing a pair of dice:

$$\Omega = \{(k, l) : k, l = 1, \dots, 6\}.$$

$N = \{1, 2\}$, Player 1 observes the first die, Player 2 observes the sum: $T_1 = \{1, 2, \dots, 6\}$, $\tau_1(\omega) = \tau_1(k, l) = k$; and $T_2 = \{2, \dots, 12\}$, $\tau_2(\omega) = \tau_2(k, l) = k + l$.

Example 1.4

Hats riddle, before the announcement. $\Omega = \{r, b\}^n$, $T_i = \{r, b\}^{n-1}$, and $\tau_i(\omega) = \omega_{-i}$.

$$(N, \Omega, (T_i)_{i \in N}, (\tau_i)_{i \in N})$$

$$N = \{A, B\}$$

Ω = each state must resolve uncertainty: - Uncertainty about color
- Color-Blindness

Subindex: 1 will denote CB from Bob

2 will denote good vision from Bob

r, p, g denote colors of cars

$$\Omega = \{\omega_{g,1}, \omega_{r,1}, \omega_{g,2}, \omega_{r,2}, \omega_p\}, \quad \text{What about } T_A, T_B, \tau_A, \tau_B$$

$$T_A = \{t_A^1, t_A^2\}; \quad \tau_A(\omega_{g,1}) = \tau_A(\omega_{g,2}) = \tau(\omega_{r,1}) = \tau(\omega_{r,2}) = t_A^1$$

$$\tau_A(\omega_p) = t_A^2$$

$$T_B = \{t_B^1, t_B^2, t_B^3, t_B^4\} \quad \tau_B(\omega_{g,1}) = \tau_B(\omega_{r,1}) = t_B^1$$

$$\tau_B(\omega_{g,2}) = t_B^2, \quad \tau_B(\omega_{r,2}) = t_B^3, \quad \tau_B(\omega_p) = t_B^4$$

$$\mathcal{P}_A = \{\{\omega_{g,1}, \omega_{g,2}, \omega_{r,1}, \omega_{r,2}\}, \{\omega_p\}\}, \quad \mathcal{P}_B = \{\{\omega_{g,1}, \omega_{r,1}\}, \{\omega_{g,2}, \omega_{r,2}\}, \{\omega_p\}\}$$

Consider a knowledge space $(N, \Omega, \{T_i\}_{i \in N}, \{\tau_i\}_{i \in N})$. Suppose $A \in 2^\Omega$ is an event.

Definition 1.5

Player i **knows A at ω** if $P_i(\omega) \subseteq A$, or equivalently if $\tau_i(\omega') = \tau_i(\omega)$ implies $\omega' \in A$.

Definition 1.6

Define $K_i A$ as the set of states of the world in which i knows A :

$$\begin{aligned} K_i A &= \{\omega : P_i(\omega) \subseteq A\} \\ &= \bigcup \{P_i(\omega) : P_i(\omega) \subseteq A\}. \end{aligned}$$

Remark: $K_i A$ is itself an event, i.e., $K_i A \in 2^\Omega$. Therefore we can talk about player j knowing if player i knows A , i.e., $K_j K_i A$. $\Rightarrow$ we can iterate the knowledge operator.

HW example continued:

$$\mathcal{P}_A = \{ \{ \omega_{g1}, \omega_{g2}, \omega_{r1}, \omega_{r2} \}, \{ \omega_p \} \}, \quad \mathcal{P}_B = \{ \{ \omega_{g1}, \omega_{r1} \}, \{ \omega_{g2} \}, \{ \omega_{r2} \}, \{ \omega_p \} \}$$

$$A = \{ \text{Akbarbek is color-blind} \} = \Omega$$

$$K_A \Omega = \Omega$$

$$\text{Does B know that A is color-blind: } K_B(K_A \Omega) = K_B \Omega = \Omega$$

$$\text{Event: "Car is Purple \& Boba is } \begin{matrix} \text{color-Blind} \\ \text{or} \\ \text{not-color-Blind} \end{matrix} = \{ \omega_p \}$$

$$K_A \{ \omega_p \} = \omega_p = \{ \omega : P_A(\omega) \subseteq \{ \omega_p \} \}$$

$$K_B(K_A \{ \omega_p \}) = K_B \{ \omega_p \} = \omega_p$$

$$\text{Is Boba color Blind: } D = \{ \omega_{g1}, \omega_{r1} \}$$

$$K_B D = D, \quad K_A(K_B D) = K_A D = \emptyset$$

$$\mathcal{P}_A = \{ \{ \omega_{g1}, \omega_{g2}, \omega_{r1}, \omega_{r2} \}, \{ \omega_{p1}, \omega_{p2} \} \} \cdot \mathcal{P}_B = \{ \{ \omega_{g1}, \omega_{r1} \}, \{ \omega_{g2}, \omega_{r2} \}, \{ \omega_{p1} \}, \{ \omega_{p2} \} \}$$

New - Model

$$\text{Bob is color-blind} = \{ \omega_{g1}, \omega_{r1}, \omega_{p1} \} = C$$

$$K_B C = C$$

$$K_A (K_B C) = K_A C = \emptyset \quad \text{"in NO state A : knows C"} \\ \text{knows that B knows C"}$$

$$\text{The car is not purple} = \{ \omega_{g1}, \omega_{g2}, \omega_{r1}, \omega_{r2} \} = D$$

$$K_A D = D, \quad K_B D = D = \{ \omega_{g1}, \omega_{r1} \} \cup \{ \omega_{g2} \} \cup \{ \omega_{r2} \}$$

Theorem 1.7 (Kripke's S5 system)

- ① $K_i\Omega = \Omega$. A player knows that some state of the world has occurred.
- ② $K_iA \cap K_iB = K_i(A \cap B)$. A player knows A and a player knows B if and only if she knows A and B .
- ③ Axiom of knowledge. $K_iA \subseteq A$. If a player knows A then A has indeed occurred.
- ④ Axiom of positive introspection. $K_iK_iA = K_iA$. If a player knows A , then she knows that she knows A .
- ⑤ Axiom of negative introspection. $(K_iA)^c = K_i((K_iA)^c)$. If a player does not know A then she knows that she does not know A .

Proof:

① By definition, it is immediate

$$\begin{aligned}\textcircled{2} \quad K_i A \cap K_i B &= \{\omega : P_i(\omega) \subseteq A\} \cap \{\omega : P_i(\omega) \subseteq B\} \\ &= \{\omega : P_i(\omega) \subseteq A, P_i(\omega) \subseteq B\} \\ &= \{\omega : P_i(\omega) \subseteq A \cap B\} \\ &= K_i(A \cap B)\end{aligned}$$

③ To show: $\omega \in K_i A \Rightarrow \omega \in A$

Pick $\omega \in K_i A \Rightarrow P_i(\omega) \subseteq A$, since $\omega \in P_i(\omega) \Rightarrow \omega \in A$
 $\Rightarrow K_i A \subseteq A$

④ We need to show 1) $K_i K_i A \subseteq K_i A$ 2) $K_i K_i A \supseteq K_i A$

1) By Axiom ③ 1) is immediate

For 2) Let $\omega \in K_i A$, T.S. $\omega \in K_i K_i A$. The fact that $\omega \in K_i A \Rightarrow P_i(\omega) \subseteq A$.

Now choose $\omega' \in P_i(\omega)$. Hence $P_i(\omega) = P_i(\omega')$, then it follows $P_i(\omega') \subseteq A$

$\omega' \in K_i A \Rightarrow P_i(\omega) \subseteq K_i A \Rightarrow K_i K_i A$

since ω' was arbitrary

⑤ Similar to ④ HW!

S5 characterizes knowledge operators

Remark

Let $L : 2^\Omega \rightarrow 2^\Omega$ be map satisfying S5. Then, there exists a type space T and a function $\tau : \Omega \rightarrow T$ such that L is equal to the associated knowledge operator given by

$$KA = \{\omega : P(\omega) \subseteq A\}$$

where $P(\omega) = \tau^{-1}(\tau(\omega))$.

S5 are not independent

Definition 1.8

An event $A \subseteq 2^\Omega$ is said to be **self-evident** to player i if $K_i A = A$. Let Σ_i be the collection of self-evident events.

Example 1.9 (Color-blindness, extended)

Recall: $\Omega = \{\omega_{g1}, \omega_{g2}, \omega_{r1}, \omega_{r2}, \omega_{p1}, \omega_{p2}\}$, $\mathcal{P}_A = \{\{\omega_{g1}, \omega_{g2}, \omega_{r1}, \omega_{r2}\}, \{\omega_{p1}, \omega_{p2}\}\}$, $\mathcal{P}_B = \{\{\omega_{g1}, \omega_{r1}\}, \{\omega_{g2}\}, \{\omega_{r2}\}, \{\omega_{p1}\}, \{\omega_{p2}\}\}$. Then
 $\Sigma_A = \{\emptyset, \{\omega_{p1}, \omega_{p2}\}, \{\omega_{g1}, \omega_{g2}, \omega_{r1}, \omega_{r2}\}, \Omega\}$,
 $\Sigma_B = \{\emptyset, \{\omega_{g2}\}, \{\omega_{p1}\}, \{\omega_{p2}\}, \{\omega_{r2}\}, \{\omega_{g1}, \omega_{r1}\}, \{\omega_{g2}, \omega_{p1}\}, \{\omega_{g2}, \omega_{p2}\}, \{\omega_{g2}, \omega_{r2}\}, \{\omega_{p1}, \omega_{p2}\}, \{\omega_{p1}, \omega_{r2}\}, \{\omega_{p2}, \omega_{r2}\}, \{\omega_{g1}, \omega_{g2}, \omega_{r1}\}, \{\omega_{g1}, \omega_{p1}, \omega_{r1}\}, \{\omega_{g1}, \omega_{p2}, \omega_{r1}\}, \{\omega_{g1}, \omega_{r1}, \omega_{r2}\}, \{\omega_{g2}, \omega_{p1}, \omega_{p2}\}, \{\omega_{g2}, \omega_{p1}, \omega_{r2}\}, \{\omega_{g2}, \omega_{p2}, \omega_{r2}\}, \{\omega_{p1}, \omega_{p2}, \omega_{r2}\}, \{\omega_{g1}, \omega_{g2}, \omega_{p1}, \omega_{r1}\}, \{\omega_{g1}, \omega_{g2}, \omega_{p2}, \omega_{r1}\}, \{\omega_{g1}, \omega_{g2}, \omega_{r1}, \omega_{r2}\}, \{\omega_{g1}, \omega_{p1}, \omega_{p2}, \omega_{r1}\}, \{\omega_{g1}, \omega_{p1}, \omega_{r1}, \omega_{r2}\}, \{\omega_{g1}, \omega_{p2}, \omega_{r1}, \omega_{r2}\}, \{\omega_{g2}, \omega_{p1}, \omega_{p2}, \omega_{r2}\}, \{\omega_{g1}, \omega_{g2}, \omega_{p1}, \omega_{p2}, \omega_{r1}\}, \{\omega_{g1}, \omega_{g2}, \omega_{p1}, \omega_{r1}, \omega_{r2}\}, \{\omega_{g1}, \omega_{g2}, \omega_{p2}, \omega_{r1}, \omega_{r2}\}, \{\omega_{g1}, \omega_{p1}, \omega_{p2}, \omega_{r1}, \omega_{r2}\}, \Omega\}$

Definition 1.10

A **field of sets** is a pair $(\Omega, \mathcal{A})$, where $\mathcal{A}$ is an **algebra over Ω** , which is a family of subsets with the properties:

A1 $\emptyset \in \mathcal{A}$ and $\Omega \in \mathcal{A}$.

A2 If $A \in \mathcal{A}$, then $A^c \in \mathcal{A}$.

A3 If $A, B \in \mathcal{A}$, then $A \cap B \in \mathcal{A}$ and $A \cup B \in \mathcal{A}$.

By induction, $\mathcal{A}$ is closed under finite intersections and finite unions.

A **σ -field** is a field equipped with a **σ -algebra**, that is, closed under countable intersections and countable unions. A **complete field** is a field equipped with an algebra closed under arbitrary intersections and unions.

Ω finite $\Rightarrow$ If $(\Omega, \mathcal{A})$ is a field it also a complete field.

Lemma 1.11

The intersection of a family of fields is a field.

The intersection of a family of σ -fields is a σ -field.

The intersection of a family of complete fields is a complete field.

Thus, given a family $\mathcal{B}$ of subsets, there is a smallest field $\mathcal{F}(\mathcal{B})$ that includes $\mathcal{B}$, a smallest σ -field $\sigma(\mathcal{B})$ that includes $\mathcal{B}$, and a smallest complete field $\mathcal{F}^(\mathcal{B})$ that includes $\mathcal{B}$.*

$$\mathcal{B} = \{ \{1\}, \{2\}, \{3,4\} \}, \quad \mathcal{F}(\mathcal{B}) = \{ \emptyset, \{1,2,3,4\}, \{1\}, \{2\}, \{3,4\}, \{1,3,4\}, \{1,2\}, \{2,3,4\} \}$$

$(\Omega, 2^\Omega)$ – Largest field of Ω

$(\Omega, \{ \emptyset, \Omega \})$ – Smallest field of Ω

Definition 1.12

An **atom** of a field is an element of the field that cannot be written as the disjoint union of two nonempty elements of the field.

Proposition 1.13

Let $\mathcal{P}$ be a partition. For any A , define $P(A) = \bigcup \{P(\omega) : \omega \in A\}$. A set A belongs to $\mathcal{F}^(\mathcal{P})$ if and only if $P(A) = A$. The set A is an atom of $\mathcal{F}^*(\mathcal{P})$ if and only if $A \in \mathcal{P}$.*

Proposition 1.14

If $\mathcal{F}$ is a complete field of subsets of Ω , the atoms of $\mathcal{F}$ constitute a partition of Ω .

Recall that Σ_i is the collection of self-evident events for player i . Let $A \in 2^\Omega$.

Claim 1.15

- ① A is self-evident if and only if it is the union of partition elements $P_i(\omega)$
- ② Σ_i is an algebra over Ω , and the field (Ω, Σ_i) coincides with $\mathcal{F}^*(\mathcal{P}_i)$, the complete field generated by $\mathcal{P}_i$.

Remark

An alternative to specifying type sets or partitions, we can start by specifying the self-evident algebras Σ_i . We can recover $P_i(\omega)$ (or T_i and τ_i) from Σ_i by

$$P_i(\omega) = \bigcap \{S \in \Sigma_i : \omega \in S\}.$$

Thus, a knowledge space can be described by $(N, \Omega, \{\Sigma_i\}_{i \in N})$.

Proposition 1.16

Let $A \in 2^\Omega$. Then $K_i A \in \Sigma_i$, and in particular

*

$$K_i A = \bigcup \{S \in \Sigma_i : S \subseteq A\}.$$

Since any event algebra is closed to unions, it follows that $K_i A$ is the largest element of Σ_i that is contained in A :

$$K_i A = \max\{S \in \Sigma_i : S \subseteq A\}.$$

* The Range of K_i is Σ_i ; for any $A \in 2^\Omega$ $K_i A \rightarrow$ a self-evident event
spits out

Definition 1.17

Let $(N, \{T_i\}_{i \in N}, \{\tau_i\}_{i \in N})$ be a finite knowledge space. An event A is said to be **common knowledge** at $\omega \in \Omega$ if for any sequence $i_1, i_2, \dots, i_k \in N$ it holds that

$$\omega \in K_{i_1} K_{i_2} \cdots K_{i_k} A$$

Definition 1.18

Let Σ, Π be two sub-algebras of some algebra. We say that Σ is a **refinement** of Π if $\Pi \subseteq \Sigma$. In this case we also say that Π is a coarsening of Σ .

Definition 1.19

The **meet** of two algebras Σ_1 and $\Sigma_2 \subseteq \Sigma$ is the finest sub-algebra of Σ that is a coarsening of each Σ_i . Their **join** is the coarsest sub-algebra of Σ that is a refinement of each Σ_i .

An alternative description of Common Knowledge events

Let $\Sigma_C = \bigcap_{i \in N} \Sigma_i$. Ex: Σ_A, Σ_B from Ahtahok & Bohn $\Sigma_C = \Sigma_A \cap \Sigma_B = \Sigma_A$

Claim 1.20

The following are equivalent:

- 1 $C \in \Sigma_C$.
- 2 $K_i C = C$ for all $i \in N$. $\Rightarrow C$ is self-evident for all i

Proof: $K_i A = A$ iff $A \in \Sigma_i$.

Hence $K_i C = C$ for all $i \Leftrightarrow C \in \Sigma_i$ for all $i \Leftrightarrow C \in \Sigma_C = \bigcap_{i=1}^n \Sigma_i$

Recall that

$$K_i A = \bigcup \{S \in \Sigma_i : S \subseteq A\} = \max\{S \in \Sigma_i : S \subseteq A\}.$$

Similarly we define:

$$K_C A = \bigcup \{S \in \Sigma_C : S \subseteq A\} = \max\{S \in \Sigma_C : S \subseteq A\}.$$

- It is useful to think about common knowledge in terms of a graph¹, which we will call G_C .
- The vertices of G_C are elements of Ω , and there is an edge between $\omega, \omega' \in \Omega$ if there exists a player i such that $P_i(\omega) = P_i(\omega')$.
- $C(\omega)$ denotes the **connected component of ω** , i.e., the maximal set of vertices in the graph for which there exists a path² from ω .

¹An undirected graph G , is a collection (V, E) where V is a set of vertices and E is a set of edges between those vertices.

²A path from ω to ω' is a sequence of edges $(\omega_1, \omega_2), (\omega_2, \omega_3), \dots, (\omega_{k-1}, \omega_k)$ such that $\omega_1 = \omega$ and $\omega_k = \omega'$.

Theorem 1.21

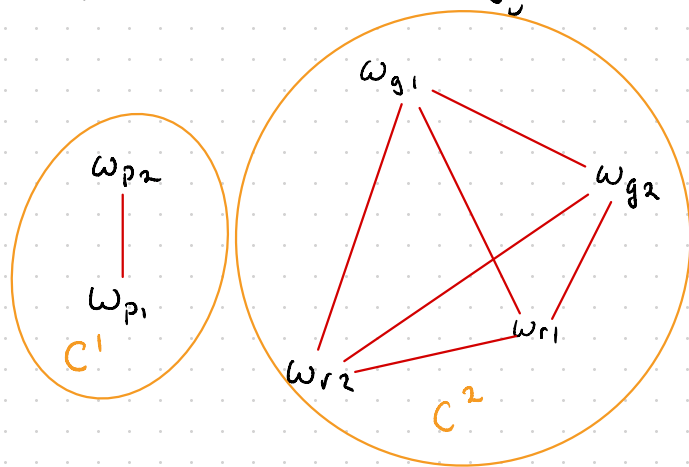
Given a finite knowledge space $(N, \Omega, \{\Sigma_i\}_{i \in N})$, a state of the world $\omega \in \Omega$ and an event A , the following are equivalent:

- ① *A is common knowledge at ω .*
- ② *$C(\omega) \subseteq A$.*
- ③ *$\omega \in K_C A$.*

Extended Cars / Color Blindness example : What events are common knowledge?

$$\mathcal{P}_A = \{ \{ \omega_{g1}, \omega_{g2}, \omega_{r1}, \omega_{r2} \}, \{ \omega_{p1}, \omega_{p2} \} \}, \mathcal{P}_B = \{ \{ \omega_{g1}, \omega_{r1} \}, \{ \omega_{g2}, \omega_{r2} \}, \{ \omega_{p1} \}, \{ \omega_{p2} \} \}$$

$G_C =$



$$C^1 = \{ \omega_{p1}, \omega_{p2} \}$$

$$C^2 = \{ \omega_{g1}, \omega_{g2}, \omega_{r1}, \omega_{r2} \}$$

C^1 and C^2 are the only connected components, thus $\Sigma_c = \{ \emptyset, \Omega, C^1, C^2 \}$

The hats riddle

Recall $\Omega = \{r, b\}^n$, $T_i = \{r, b\}^{n-1}$, and $\tau_i(\omega) = \omega_{-i}$. The information partition functions are $P_i(\omega) = \{(\omega_{-i}, r), (\omega_{-i}, b)\}$. Thus the set of edges of G_C is

$$E = \{P_i(\omega)\}_{i,\omega} = \{(\omega_{-i}, r), (\omega_{-i}, b)\} : \omega_{-i} \in \{r, b\}^{n-1}\}$$

Claim 1.22

$$\Sigma_C = \{\emptyset, \Omega\}.$$

Proof: It is easy to see that for every $\omega, \omega' \in \Omega$ there is a sequence of edges (*corresponding to the coordinates in which they differ) that forms a path from ω to ω' in G_C . Thus $C(\omega) = \Omega$ and the only CK events are Ω and $\emptyset$.

* For example, $n=3$, One path from (r, r, r) to (b, b, b) is: $(r, r, r) \rightarrow (b, r, r) \rightarrow (b, b, r) \rightarrow (b, b, b)$
